B.Sc (Hons) Information Technology
IE3041 - Data Management and Business Intelligence
2025 Jan - June
Assignment

Instructions:

You have **1 hour and 30 minutes** to complete this test and submit your answers.

- **Dataset:** Review the "**Seasonal_Offer_Sales_data"** dataset carefully before answering the questions.
- **Answers:** For each question, take a screenshot of your answer in Power BI and insert it into a Word document.
- **File Naming:** Save both your Power BI file and Word document using your **student ID** as the file name. Convert the Word document to a **PDF** before submission.
- **Submission:** Ensure that each question includes a corresponding screenshot. If a screenshot is missing, **half marks** will be given for that question.
- **Submission Format:** Submit your work by uploading the Power BI file and the PDF file separately to their respective submission links.
- **Device Restrictions:** Students are **not allowed** to use any mobile phones or other smart devices during the test. You are only permitted to use **Power BI, Word, and the SLIIT Courseweb**.

ANY VIOLATION OF THESE INSTRUCTIONS WILL RESULT IN A **ZERO** FOR THE TEST.

Questions:

Load the dataset “Seasonal_Offer_Sales_data” into Power BI.

1. Add an Index Column starting from 1 to the dataset and position it as the first column.
2. Create a new calculated column called "Seasonal Discount" that calculates the difference between the "Base Price" and "Discounted Price."
3. Create a new column called "Offer Type" to categorize offers based on the "Discount Percentage":

   Low Offer for discounts below 10%

   Medium Offer for discounts between 10% and 30%

   High Offer for discounts above 30%
4. Create a calculated column "Revenue per Unit" that shows how much revenue was made per unit sold.
5. Create a measure to calculate the total number of units sold across all seasons (Include the DAX formula for this measure.)
6. In a chart showing the total revenue generated by each product category, also display their average revenues.
7. Create a pie chart showing the distribution of products by "Offer Type" (Low, Medium, High).
8. Using the "Discount Percentage" column, create a visual that shows the average discount across different product categories.
9. Create a Card visual that shows the Total Revenue during the "Winter" season, based on the filter.
10. Perform a Month-Wise Revenue Forecasting for the Next Year.

Marking Guide

Criteria	Marks
Get data into Power BI	1
Index column	1
Create column – Seasonal Discount	5
Create column – Offer Type	5
Create column – Revenue per Unit	5
Create measure – Total Units Sold	5
Create a Column Chart: Total Revenue per Product Category, with Average revenues	5
Create a Pie Chart: Distribution by "Offer Type"	5
Create a visual: Average Discount % per Product Category	6
Create a Card Visual showing Total Revenue during the Winter season	5
Perform Month-wise Revenue Forecasting for the Next Year	7
Total	50